8 **(a)** **(i)** The root mean square value of an alternating current that passes through a <u>resistor</u> is the value of an equivalent steady direct current that would produce the same <u>rate of heating</u> in the resistor as the alternating current .

Pure definition only 1 mark [B1].
Contextualising [B1]

(ii)

1.

$$V_o = \sqrt{2} \times V_{r.m.s}$$
$$= \sqrt{2} \times 230 = 325 \text{ V } [A1]$$

2.

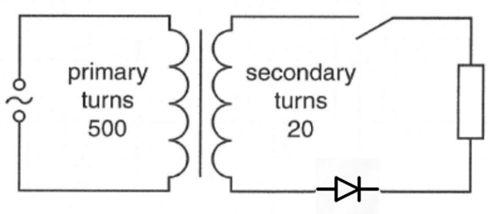
$$\omega = 2\pi f = 2\pi \times (\frac{1}{20 \times 10^{-3}})[C1]$$
$$= 314 \; rad \; s^{-1}[A1]$$

3. $V = 325\sin(314t)$ [A1]

(iii)

$$P_{mean} = \frac{V^2_{r.m.s}}{R} = \frac{(\frac{V_o}{\sqrt{2}})^2}{R}[M1]$$
$$= \frac{V_o^{\,2}}{2R}[A0] = \frac{P_o}{2}$$

(iv)



[B1] for correct set up and symbol used

A <u>step down</u> transformer can be used to reduce the voltage of the power supply [B1].

A diode can then be used to convert the AC to DC (or rectification mentioned) [B1].

(b) **(i)** Current flow from B to A if circuit is closed. As coil is taken as battery, current flow from –ve to +ve.Thus, B is labelled "-" and A "+"[B1]

(ii) **correct shape** [M1] – accept both positive or negative cos or sine graph, 2 complete cycles within 1.0 s with period labelled as 0.50 s [A1]

(iii) Since the velocity varies sinusoidally with time, the rate of magnetic flux linkage also varies sinusoidally with time. [M1]

Since induced emf is proportional to the rate of change of magnetic flux linkage. Thus, by Faraday's law, emf varies sinusoidally with time [A1].

(c) **(i)** As the circuit is now closed, induced current will flow through the coil. [B1] By Lenz's law, the induced current will flow in such a way to **produce an effect** to oppose the change in magnetic flux linkage experienced by the coil [M1]. This will create a damping effect on the magnet and it will eventually stop oscillating. [A1]

Alternative answer: using COE to explain: As the circuit is now closed, induced current will flow through the coil [B1].The mechanical energy of the magnet is converted to electrical energy of coil [M1], hence the amplitude of oscillation decreases and will eventually becomes zero [A1].

(ii) Shape- Cosine square or sine square graph [M1] with decreasing amplitude and constant (or slightly longer) period [A1]

Dependent on answer given in b(ii)